

NATURAL LANGUAGE PROCESSING · MS DATA SCIENCE · FINAL PROJECT

ClearClause

AI-Powered Contract Risk Analyzer
for Freelancers & Professionals

Shehroz Ali

MSDSF25M012

Arslan Ahmed

MSDSF25M001

1 Group Members

Shehroz Ali

MSDSF25M012

NLP Pipeline · Model Integration · Evaluation

Arslan Ahmed

MSDSF25M001

System Design · UI / Demo · Business Analysis

2 Project Description

2.1 The Problem

Legal documents — employment contracts, NDAs, freelance service agreements, lease agreements, and terms-of-service policies — are deliberately complex. The average employment contract runs to **4,500+ words**; a Fiverr Terms of Service exceeds **10,000 words**. Yet the consequences of misunderstanding them are severe: lost intellectual property rights, unexpected non-compete restrictions, automatic subscription renewals, and personal liability for company debts.

Pakistan is ranked **4th globally in freelancers**, with over **4 million registered freelancers** generating **\$1.5 billion+ annually** via Fiverr, Upwork, and Freelancer.com. These individuals regularly sign client contracts drafted by international lawyers — yet *no affordable, accessible tool exists to help them understand what they are agreeing to*. This creates a profound **information asymmetry**: corporations draft with their legal teams; ordinary people sign with their fingers crossed.

2.2 The Solution

ClearClause is an intelligent, NLP-powered contract analysis system that transforms complex legal language into immediate, actionable understanding — in under 60 seconds.

The system delivers five core capabilities:

1. **Clause Extraction & Risk Classification** — Identifies and flags dangerous, unusual, or one-sided clauses (e.g., unlimited liability, automatic renewal, unilateral termination)
2. **Plain-English Simplification** — Converts every flagged clause from legalese into 1–2 clear, readable sentences
3. **Document Summarization** — Produces a structured "Key Points" report covering obligations, rights, risks, and deadlines
4. **Conversational Q&A** — Answers user questions via an AI chatbot grounded exclusively in the uploaded contract
5. **Risk Scoring** — Assigns an overall 0–100 fairness/risk score displayed as a color-coded gauge

3 Technical Perspective

3.1 System Architecture



3.2 Core NLP Modules

Module 1 — Document Ingestion

- **Input:** PDF or DOCX file uploaded by the user
- **Libraries:** `pdfplumber` (PDF), `python-docx` (Word)
- **Process:** Text extraction → whitespace normalization → paragraph-level chunking
- **Output:** Clean, structured text corpus ready for NLP processing

Module 2 — Clause Extraction & Risk Classification (Core NLP)

This is the primary NLP contribution, combining **Extractive Question Answering** with **Multi-label Text Classification** on the legal domain.

- **Model:** `tomasonjo/deberta-v3-base-cuad` — DeBERTa-v3 fine-tuned on the CUAD dataset (13,500+ expert-annotated legal clauses, **41 risk categories**)

- **Approach:** Each clause category is posed as a QA query; the model performs span extraction — no additional training required (transfer learning)
- **Why DeBERTa-v3:** Disentangled attention mechanism is particularly effective for legal text; outperforms BERT and RoBERTa on QA benchmarks

CLAUSE TYPE	RISK LEVEL	WHY IT MATTERS
Unlimited / All-or-Nothing Liability	 HIGH	Exposes individual to unlimited financial loss
IP Ownership / Work-for-Hire	 HIGH	May strip creator of ownership of their own work
Automatic Renewal	 HIGH	Hidden recurring commitment, difficult to exit
Unilateral Termination	 HIGH	Client can terminate contract without stated cause
Non-Compete / Non-Solicitation	 MED	Restricts future employment opportunities
Indemnification	 MED	May require paying the client's legal costs
Governing Law / Jurisdiction	 MED	Disputes resolved in a foreign court
Arbitration Clause	 MED	Waives the right to sue in normal court
Payment Terms	 LOW	Standard, expected commercial clause
Effective Date	 LOW	Informational, no risk to signer

Module 3 — Plain-English Simplification

- **Technique:** Instruction-tuned LLM via Groq API (`llama3-8b-8192`)
- **Prompt:** *"You are a legal plain-language translator. Rewrite the following clause in 2 clear sentences a non-lawyer can understand. Preserve all legal meaning."*
- **Why Groq:** Free tier, 700+ tokens/sec, sufficient quality for clause-level translation with sub-second latency
- **Output:** Plain-English version of every HIGH and MEDIUM risk clause

Module 4 — Conversational Q&A via RAG

- **Architecture:** Retrieval-Augmented Generation (RAG) pipeline
- Document chunks embedded via `sentence-transformers` (`all-MiniLM-L6-v2`)
- Embeddings stored in a **FAISS** in-memory vector index (no database needed)
- On user query → semantic top-k search → relevant chunks retrieved → Groq LLM generates grounded answer
- **Framework:** LangChain `RetrievalQA` chain
- **Example queries:** *"Can the client fire me without reason?" · "Who owns the code I write?" · "Can I work for other clients?"*

Module 5 — Document Summarization

- **Technique:** Structured prompt-based abstractive summarization via Groq LLM
- **Fixed output schema:** Key Obligations · Rights Granted · Key Risks · Important Dates/Deadlines
- **Length:** Under 300 words — readable at a glance

Risk Score = min(100, (HIGH_count × 30 + MED_count × 15 + LOW_count × 5) / N_expected × 100)

● 0–33

● 34–66

● 67–100

● Safe

● Review Carefully

● High Risk

3.3 Datasets

DATASET	PURPOSE	STATUS
CUAD (Contract Understanding Atticus Dataset)	Pre-trained QA model — 13,500+ annotations, 41 categories	✓ Available
TOS;DR (Terms of Service; Didn't Read)	Evaluation of simplification quality	✓ Available
LEDGAR (SEC legal provisions)	Additional clause variety for testing	✓ Available
Sample Freelance Contracts (manual)	Demo and end-to-end evaluation	📋 To collect

3.4 Technology Stack

LAYER	TECHNOLOGY	JUSTIFICATION
NLP Backbone	<code>deberta-v3-base-cuad</code>	SOTA on legal clause QA (CUAD benchmark)
LLM API	Groq (<code>llama3-8b-8192</code>)	Free tier, 700 tok/s, low latency
Embeddings	<code>sentence-transformers</code>	Efficient semantic vector search
Vector Store	FAISS	In-memory, zero infrastructure needed
RAG Framework	LangChain	Industry standard, well-documented
Document Parsing	pdfplumber, python-docx	Handles PDF and Word formats
Web Interface	Streamlit	Python-native, rapid prototyping
Visualization	Plotly	Interactive risk gauge chart
Language	Python 3.10+	Ecosystem & team proficiency

3.5 Evaluation Metrics

MODULE	METRIC	BASELINE / TARGET
Clause Extraction	F1-Score, Exact Match (EM)	CUAD benchmark: 87%+ F1
Risk Classification	Accuracy, Macro-F1	Evaluated on held-out CUAD split
Summarization	ROUGE-1/2/L	vs. human-written reference summaries
Simplification	Flesch-Kincaid readability improvement	Original vs. simplified clause reading level
Q&A Faithfulness	BERTScore	Answer grounded in source document

3.6 4-Day Prototype Development Plan

DAY	SHEHROZ ALI	ARSLAN AHMED
1	Document ingestion pipeline + CUAD model integration for clause extraction	Project repo setup, Streamlit boilerplate, Groq API integration
2	RAG pipeline: LangChain + FAISS + sentence-transformers	UI: file upload, clause display with color-coded risk badges
3	Risk scoring engine + end-to-end integration testing	UI: risk gauge chart, summary panel, Q&A chat interface
4	Evaluation on 10+ sample contracts + metrics report	Demo recording, documentation, final polish

4 Business Perspective

4.1 Problem Statement

Legal literacy is a systemic global failure. The information asymmetry between those who *draft* contracts (corporations, their lawyers) and those who *sign* them (individuals, freelancers, employees) consistently causes harm to the less powerful party. Consider what routinely happens when people sign without truly understanding:

- A freelancer signs an IP transfer clause and unknowingly loses all rights to the code they wrote
- An employee signs a non-compete that prevents them from working in their field for two years after leaving
- A small business owner signs an indemnification clause and becomes personally liable for their client's legal costs

Pakistan's growing freelancer class faces this daily — dealing with international contracts under significant legal *and* linguistic disadvantage, with zero affordable tools to help them.

4.2 Target Market

SEGMENT	SIZE	PAIN POINT
Pakistani freelancers (primary)	4M+ registered	Client contracts in English legal language
Global gig workers (Fiverr/Upwork)	73M+ in US alone	IP clauses, payment disputes, non-competes
Fresh graduates / new employees	Millions annually	Employment contracts, probation terms
Small businesses & startups	33M SMBs (US)	Vendor agreements, SaaS subscriptions
Renters	Universal	Lease auto-renewal and penalty clauses

4.3 Business Opportunity

The global **LegalTech market** is valued at **\$30 billion (2023)** and projected to reach **\$88 billion by 2030** (CAGR 16.7%). AI-powered document review is the highest-growth subsegment. Pakistan alone represents a completely **untapped first-mover opportunity** — 4M+ freelancers, \$1.5B/yr earnings, zero AI tools targeting them.

4.4 Value Proposition

"ClearClause gives every freelancer, employee, and small business the power to truly understand what they are signing — in plain English, in under 60 seconds, at a fraction of the cost of a lawyer."

TRADITIONAL APPROACH	CLEARCLAUSE
\$200–\$500 per hour (lawyer)	\$9.99/month subscription
2–5 business days turnaround	Instant — under 60 seconds
Only viable for large companies	Accessible to any individual
Generic legal advice	Document-specific, clause-level insight
No grounding in your actual contract	All answers sourced from your document

4.5 Revenue Model

TIER	PRICE	FEATURES
Free	\$0 / month	3 documents/month, basic summary only
Pro	\$9.99 / month	Unlimited documents, full risk analysis, Q&A, risk score

TIER	PRICE	FEATURES
Business	\$49 / month	Team access, API, bulk processing, branded reports
Enterprise	Custom	White-label, CRM integrations, dedicated support

Additional Revenue Streams:

- **B2B API Licensing** — Integration into Fiverr, Upwork, LinkedIn for in-platform contract review
- **Legal Referral Marketplace** — Connect high-risk users to vetted lawyers (10–15% referral fee)
- **White-label for HR Platforms** — Companies embed ClearClause for employee onboarding workflows

4.6 Competitive Landscape

COMPETITOR	LIMITATION	OUR ADVANTAGE
Harvey AI	\$40K+/year, enterprise only	Consumer-grade pricing, accessible to all
LawGeex	Enterprise, not for individuals	Individual-first product design
Kira Systems	M&A focus, complex interface	Any contract type, simple UX
ChatGPT (generic)	No legal specialization, hallucination risk	Purpose-built legal NLP, source-grounded
Manual reading	Time-consuming, error-prone	60-second automated analysis

Key Differentiator: ClearClause is the only product combining legal-domain NLP (CUAD-trained DeBERTa-v3), RAG-based document Q&A, and clause risk scoring in a single affordable interface — with specific focus on Pakistan's freelancer economy, a market currently entirely underserved by existing LegalTech products.

4.7 Go-to-Market Strategy

Phase 1
Months 1–3

Launch: Free-tier web app targeting Pakistani and global freelancers. Distribution via Reddit (r/freelance, r/Pakistan), Product Hunt, Fiverr community forums, LinkedIn.

Phase 2
Months 4–9

Growth: Pro subscription launch, content marketing ("*5 Dangerous Clauses in Your Freelance Contract*"), influencer partnerships with Pakistani tech/freelancing YouTubers.

Phase 3
Months 9–18

B2B Expansion: API product for freelance platforms and HR software. Enterprise white-label deals in real estate, staffing, and legal services.

4.8 Social Impact

Beyond commercial opportunity, ClearClause addresses a genuine **access-to-justice problem**. Legal document opacity disproportionately harms freelancers who cannot afford legal advice, first-generation professionals unfamiliar with corporate norms, individuals dealing with contracts in a second language, and residents of developing economies signing

contracts from wealthier legal jurisdictions. By democratizing legal comprehension, ClearClause provides individuals the same informational advantage traditionally held only by corporations and their legal teams.

5 Project Summary

Project Name	ClearClause
Group Members	Shehroz Ali (MSDSF25M012) · Arslan Ahmed (MSDSF25M001)
Core NLP Tasks	Extractive QA · Multi-label Risk Classification · Text Simplification · Abstractive Summarization · RAG-based Q&A
Key Models	DeBERTa-v3-CUAD · sentence-transformers · Groq LLaMA3 · FAISS
Primary Dataset	CUAD — 13,500+ annotated legal clauses, 41 categories
Deliverable	Streamlit app: upload contract → instant risk report + Q&A chatbot
Timeline	4-day prototype sprint (no from-scratch training required)
Primary Market	Pakistan freelancer economy (\$1.5B/yr) → Global gig economy (1.57B people)
Business Model	Freemium SaaS · B2B API · Legal referral marketplace
Market Size	\$30B → \$88B LegalTech market by 2030 (CAGR 16.7%)